

Smart IoT Health & Fall-Detection Bio-Band

On-Device Edge Analytics & Decentralized Captive Medical Portal

Category: Electronics (Sasthramela)	Microcontroller: ESP32-C3 RISC-V (160MHz)
Primary Sensors: MAX30102, MPU6050, DS18B20, GSR	Display & Controls: 0.96" OLED + 2 Tactile Buttons

1. Executive Summary & Abstract

Geriatric individuals and solo workers face severe risks from sudden cardiac anomalies, hypoxia, psychological stress spikes, and mechanical falls. Conventional IoT wearables rely on external cellular towers or cloud servers, failing during network outages or rural emergencies. This project introduces a fully decentralized, self-contained wearable bio-band using the **ESP32-C3 RISC-V SoC**. It performs on-device biosignal filtering, multi-screen OLED navigation via two hardware tactile buttons, and hosts a standalone Wi-Fi Captive Portal (192.168.4.1) requiring zero mobile apps for immediate paramedic triage.

2. Scientific Principles & Mathematical Formulations

A. Optical Photoplethysmography (PPG) — MAX30102

Dual-wavelength LEDs (Red 660nm, IR 880nm) measure differential absorption across pulsing arterial blood vessels. SpO2 is calculated using the ratio of ratios: $R = (AC_{red} / DC_{red}) / (AC_{ir} / DC_{ir})$, yielding $SpO2 = 110 - 25 \times R$ (%).

B. Galvanic Skin Response (GSR) / Electrodermal Activity

Measures sympathetic autonomic nervous arousal via eccrine sweat gland conductance: $R_{skin} = R_{ref} \times (V_{in} - V_{out}) / V_{out}$ (kΩ) and Conductance $G = 1000 / R_{skin}$ (μS).

C. Dual-Threshold Fall Detection — MPU6050

Total 3D Acceleration Vector Magnitude $a_{total} = \sqrt{ax^2 + ay^2 + az^2}$. The algorithm detects free-fall weightlessness ($a < 0.45g$ for >60ms) followed by ground impact ($a > 2.60g$ within 600ms) and postural alteration.

3. Hardware Interconnection & Pin Mapping

Component	Module Pin	ESP32-C3 Pin	Protocol / Description
0.96" I2C OLED	SDA / SCL	GPIO 4 / GPIO 5	Shared I2C Bus (0x3C), 3.3V
MAX30102 PPG	SDA / SCL	GPIO 4 / GPIO 5	Shared I2C Bus (0x57), 3.3V
MPU6050 IMU	SDA / SCL	GPIO 4 / GPIO 5	Shared I2C Bus (0x68), 3.3V
DS18B20 Temp	DQ (Data)	GPIO 2	OneWire Bus (4.7k Ω ; Pull-up)
GSR Electrodes	Analog Out	GPIO 0	ADC1_CH0 (12-bit Analog)
Piezo Buzzer	Positive (+)	GPIO 3	High-Decibel Pulse Siren
Button 1 (Scroll)	Pin 1	GPIO 9	Internal Pullup (Active LOW)
Button 2 (Select)	Pin 1	GPIO 8	Internal Pullup (Active LOW)
Battery Divider	Analog Mid	GPIO 1	ADC1_CH1 (100k/100k Divider)

4. 0.96" Monochrome OLED Display & Dual-Button UI

The interface incorporates two physical buttons directly below the 128x64 display: **Button 1 (Scroll)** cycles through 5 diagnostic pages, and **Button 2 (Select)** acknowledges alarms or toggles the buzzer.

Page #	Screen Title	Parameters Displayed	Visual Elements
0	Overview Dashboard	BPM, SpO2, Body Temp C/F, GSR, Motion	Status bar + Battery icon
1	Cardiac Monitor	Large Heart Rate BPM, SpO2 %	Live ECG Pulse Waveform
2	Stress & Thermal	GSR Skin $k\Omega$, μS , Body Temp	Dynamic Stress Progress Bar
3	Motion & Fall	Total G, Pitch/Roll angles, Safety State	Inverted Fall Alarm Banner
4	Wi-Fi Hotspot	SSID, Captive IP 192.168.4.1, Clients	Buzzer Mute status indicator

5. Bill of Materials (BOM) & Budget

SI No	Component Specification	Qty	Cost (INR)
1	ESP32-C3 RISC-V SoC Development Board	1	Rs. 290
2	0.96" Monochrome I2C SSD1306 OLED (128x64)	1	Rs. 160
3	MAX30102 PPG Optical Heart Rate & SpO2 Module	1	Rs. 220
4	MPU6050 6-Axis Accelerometer & Gyroscope	1	Rs. 140
5	DS18B20 High-Precision Body Temp Sensor	1	Rs. 90
6	GSR Skin Conductance Sensor + Finger Straps	1	Rs. 210
7	3.7V 500mAh LiPo Battery + TP4056 USB-C Charger	1	Rs. 180
8	Tactile Push Switches, Active Buzzer & Resistors	1 set	Rs. 30
9	Enclosure, Casing & Wrist Straps	1	Rs. 80
TOTAL PROJECT COST			Rs. 1,400 INR

6. Kerala Sasthramela Viva Voce & Technical Defense

Q1: What prevents false alarms during vigorous movements like clapping or jogging?

Defense: The algorithm implements strict two-tier time-domain hysteresis: a freefall weightlessness state ($<0.45g$) must precede an impact spike ($>2.60g$) within a 600ms window, followed by post-fall orientation rest.

Q2: How does the Captive Portal operate without an active internet connection?

Defense: The ESP32-C3 acts as an independent Wi-Fi SoftAP and runs an internal DNS Server on Port 53. All DNS lookups from connected smartphones are redirected to 192.168.4.1, automatically opening the medical dashboard in the OS default captive browser.